



Fig. 3: Writing current waveforms

writing individual cells. There is a cell selector device placed either above or below each memory cell.

The cell selector, which can be either a p-n-junction diode, a bipolar or field-effect transistor, or some other suitable device, ensures that only the addressed memory cell is selected. The selector device also provides heating current to the cell for changing its state, as required during a write operation. The cell selector must be able to provide the needed current and must have low leakage itself to reduce the overall power consumption.

Fig. 2 shows the structure of a PCM cell in set and reset states. It is advantageous to reduce the contact area between the TiN heater and the phase-change material. This increases the rate of heating by reducing thermal time constants and also the value of heating current needed to change the material's phase.

In order to program a cell into 'set' (binary '1') state, assuming it was previously in 'reset' (binary '0') state, a low current is passed for a few tens of nanoseconds through the cell. The TiN heater heats up the phase-change material to a temperature above its crystallisation temperature (around 200°C) but well below the melting temperature (around 600°C). This causes the atoms of the glassy material to move and fall into energetically more favourable periodic positions. The material thus transforms to its crystalline phase that represents binary '1.' This process transforms only a part of the entire material volume but the change

is reflected in the increased amount of crystalline material after the annealing process, as seen in Fig. 2.

In order to go from the 'set' state to the 'reset' state, on the other hand, a short-duration large current pulse is passed through the cell, which heats above the melting point of the phase-change material and then cools down very quickly. The rapid cooling does not allow the atoms of the phase-change material to fall into an ordered arrangement, resulting in an amorphous glassy state that represents binary '0.' This is how the cell is written to. The writing current waveforms are shown in Fig. 3.

In order to read the state of a memory cell, a small constant current is passed through the cell and the resulting voltage across it is measured. This reveals the resistance of the cell with a low resistance standing for binary '1' and a high resistance indicating binary '0.' The read current is too low to disturb the material state of the cell. The several hundreds of microamps required to change the state of a memory cell is a continuing concern and a lot of current development effort is being directed to reduce the power consumption of the PCM modules.

Once a cell has been programmed into its reset state by transforming the phase-change material to its amorphous state, the value of its resistance can steadily increase over time. This resistance drift was a big problem in the early days of PCM devices. Researchers at IBM research labs solved this problem by coming up with al-

gorithms that measure the resistance drift of a block of contiguous cells and extract correct bit information by statistical sampling. Fig. 4 shows an experimental PCM device from IBM research.

More bang for the buck

In recent years, two principal strategies have been explored to increase the memory capacity of the PCM chips. One is to simply stack two or more layers of phase-change cell arrays to build a multilayer memory chip. Memory cells are arranged in two-dimensional arrays with cell selectors connected to word and bit lines in a cross-point switch arrangement. Although a feasible concept, this architecture has not been implemented as a working device yet.

The other, more viable route is to make use of the large resistance interval between amorphous and crystalline forms of phase-change materials by defining intermediate resistance levels as representing various bit combinations. Thus, in a four-level system, four resistance intervals can be used to represent 00, 01, 10 and 11. A byte, such as 10011010, can then be stored in four cells instead of eight, effectively doubling the memory capacity. More fine-grained resistance levels can be established by injecting carefully-controlled current pulses to heat and program the phase-change material in such a manner that a selected fraction of it changes its phase.

Researchers have shown that it is possible to make a 32-level phase-change memory system where each



Fig. 4: An experimental PCM device from IBM Research